Detector-Plane Causality in Quantum Measurement

A Benchmark Framework for Measurement Integrity and Control

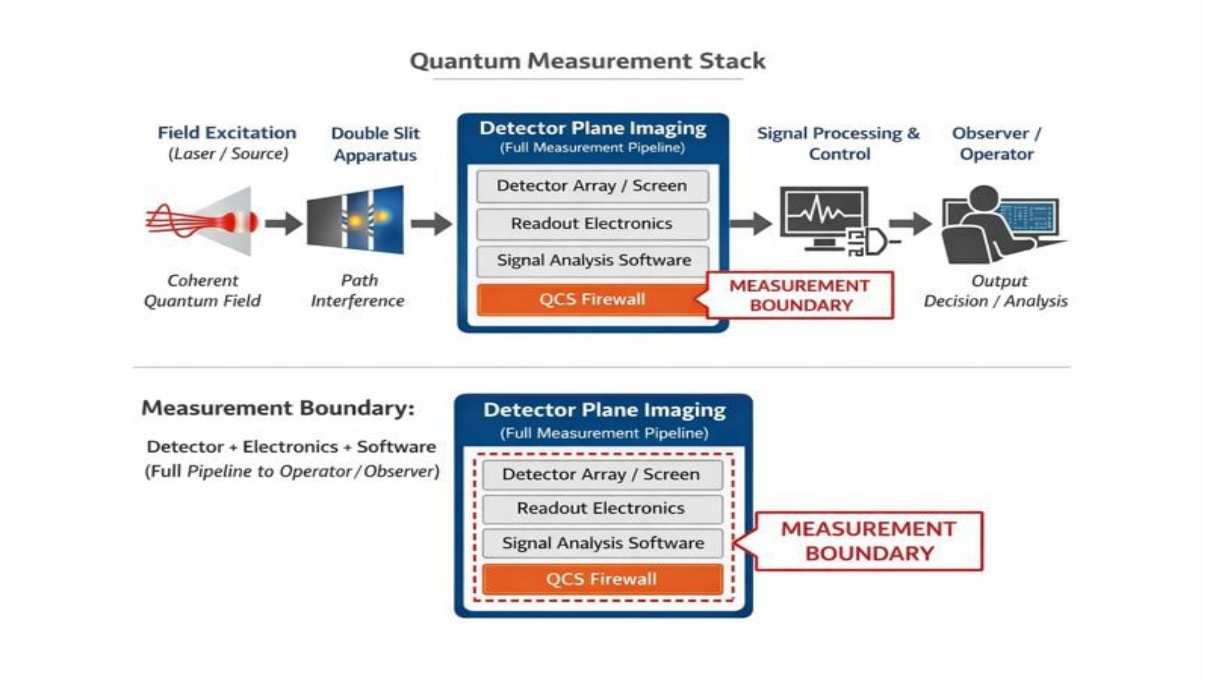

Figure 1. Quantum Measurement Stack illustrating detector-plane imaging as the effective measurement boundary.

This handout summarizes the framework proposed in **Detector-Plane Causality in Quantum Measurement**. The work treats quantum measurement as an instrumentation and signal-processing problem and introduces the **Quantum Measurement Stack (QMS)** together with the **QMCTB-01 benchmark** for evaluating detector-plane coherence preservation.

Causal Measurement Chain

Field Excitation → Imprint Formation → Detector-Plane Processing → Data Acquisition → Diagnostics and Control → Observable Measurement

Quantum Measurement Stack (QMS)

Layer	Function
Field Excitation and Imprint Formation	Quantum fields generate particle excitations and interference structures.
Detector-Plane Imaging	Sensors convert optical or particle signals into electronic signals.
Data Acquisition (DAQ)	Electronics digitize signals into measurable data.
Quantum Diagnostics	Measurement fidelity and coherence preservation are evaluated.
Quantum Control System (QCS)	Experimental parameters are adjusted to stabilize measurement integrity.

Detector Plane

The detector plane defines the effective measurement boundary and includes detector materials, analog front-end electronics, digitization hardware, firmware, and software readout. Any correlations lost at this stage are permanently removed from the measurement record.

QMCTB-01 Benchmark

Detector Regime	Observed Outcome
Intensity-Only Detection	Phase correlations discarded; no stable interference fringes appear.
Correlation-Preserving Detection	Phase correlations retained; stable high-contrast fringes appear.

Key Implications

• Irreversible information loss is localized at the detector plane.
• Interference visibility depends on detector-plane coherence fidelity.
• Measurement integrity can be treated as a controllable hardware parameter.

Research Context

The detector-plane causality framework provides a measurement-science perspective applicable across quantum optics instrumentation, single-photon detector systems, solid-state sensors, and hybrid quantum measurement platforms. The QMCTB-01 benchmark enables reproducible evaluation of detector coherence fidelity across experimental platforms.